HEMODYNAMIC DISTURBANCES 1

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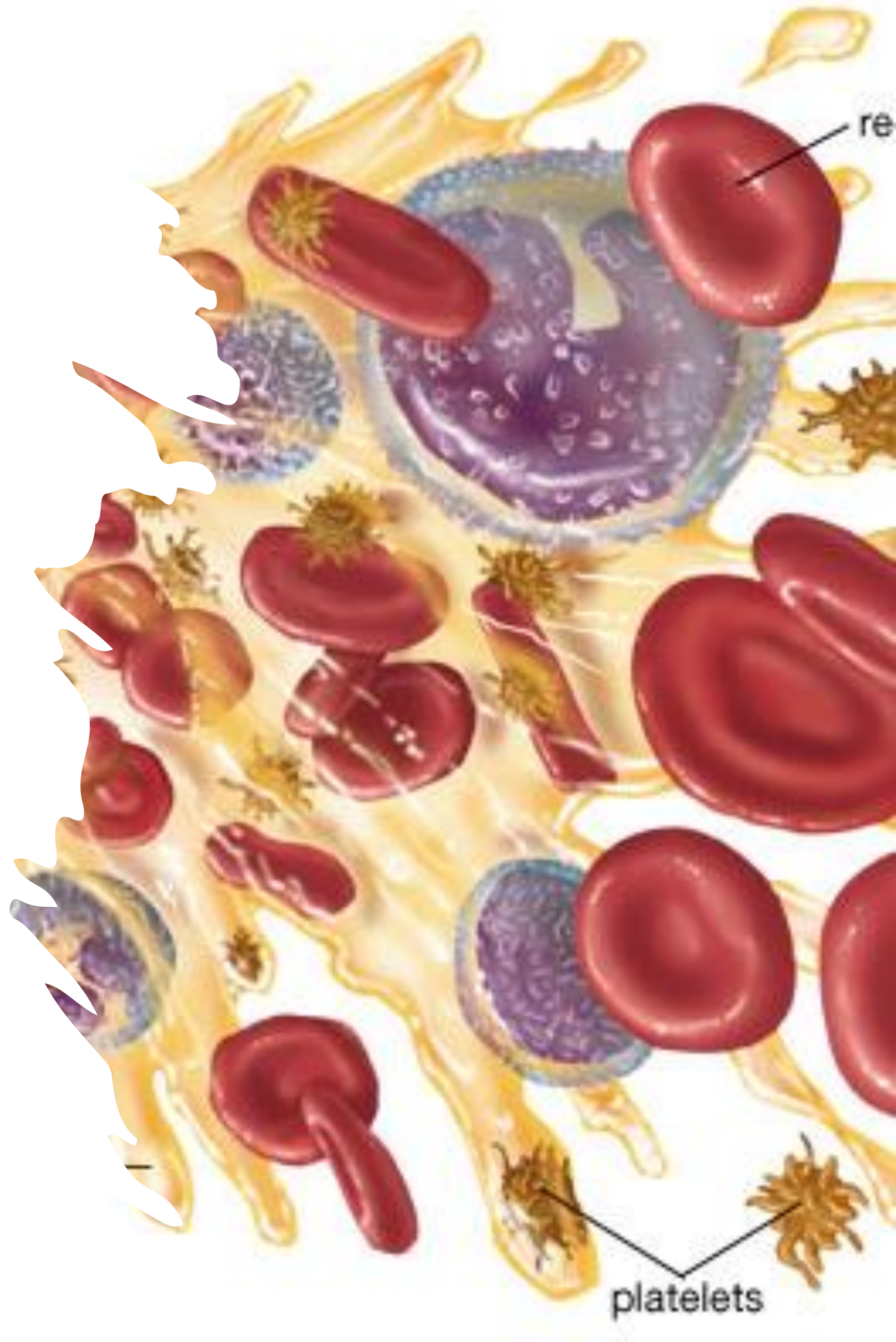


Ischemia

- Def: A reduction of arterial blood supply to a tissue

Acute ischemia
- ❖ Sudden
- ❖ Complete

Efficient collaterals

Minimal or significant effects

Poor collaterals

Ischemic necrosis:
- ❖ Infarction
- ❖ Gangrene

Chronic ischemia
- ❖ Gradual
- ❖ Incomplete

Efficient collaterals

No effects

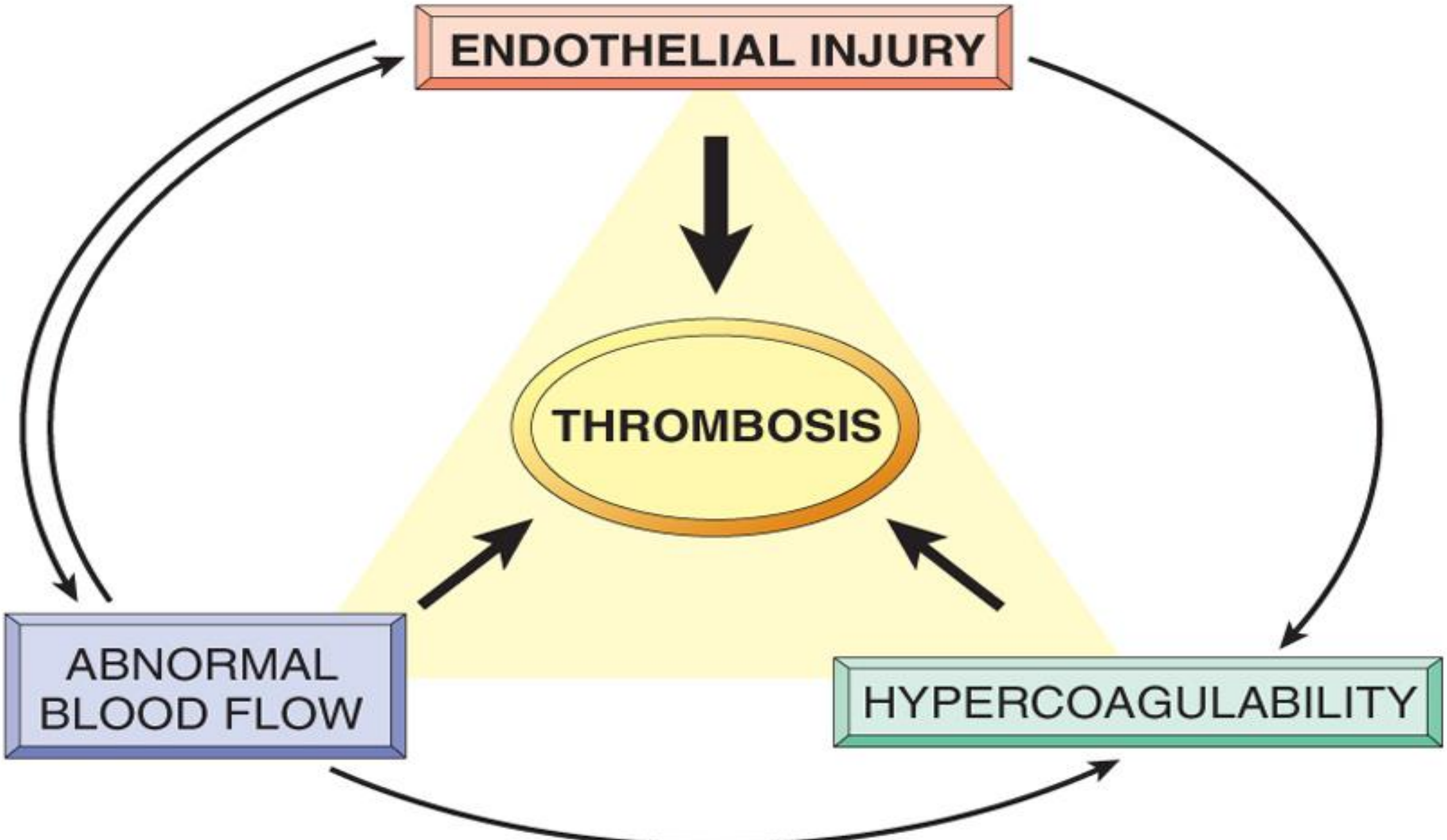
Poor collaterals

- ❖ Pain on effort
- ❖ patchy fibrosis

THROMBOSIS

- ***Definition:*** is the formation of a ***thrombus***
- Thrombus: is a compact mass formed of the elements of ***circulating*** blood ***inside*** the cardiovascular system during ***life.***

Causes of thrombosis
"Virchow's triad"



Endothelial cell injury

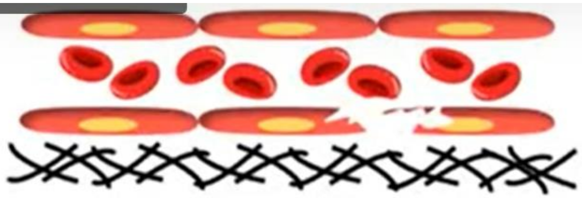
- Trauma
- Inflammation
- Atherosclerosis

Abnormal blood flow

- Stasis (prolonged immobilizatio
- Turbulence = unsteady movement(aneurysm)

Hypercoagulability of blood

- Postoperative
- Neoplasia
- Pregnancy
- Oral contraceptive pills



Endothelial injury

↓

Exposure of the sub endothelium

↓

Adhesion of platelets

↓

Fibrin deposition

↓

Entrapment of other cells (RBC, WBC)

↓

Turbulence with more platelet adhesion → Thrombus of alternating layers of platelets, fibrin

Dysfunctional endothelial cells produce :

More procoagulant factors

Less anti coagulant factors

↓

THROMBOSIS

Hypercoagulability

Less frequent contributor for thrombosis

Defined as :- Any alteration of the coagulation pathways that predisposes to thrombosis

Genetic (primary) causes

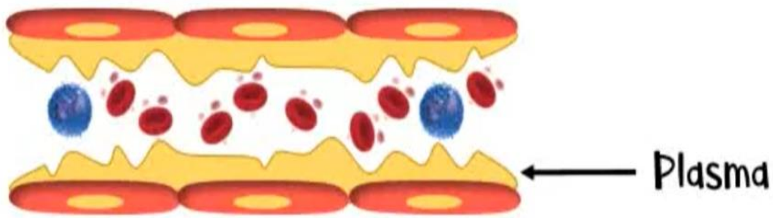
- Factor V Leiden mutation
- Prothrombin mutation
- Sticky platelet syndrome
- Elevated factor VIII, IX, XI or fibrinogen

Acquired (secondary) causes

- Prolonged bed rest or immobilization
- Tissue injury (surgery, fracture, burn)
- Myocardial infarction
- Cancer
- Disseminated intravascular coagulation (DIC)
- Heparin - induced thrombocytopenia
- Antiphospholipid antibody syndrome

Alterations in the normal blood flow

1. Turbulence
2. Stasis



Laminar flow

Prevents platelets coming in contact with the endothelium

Turbulent flow ⇨ No separation between plasma & blood cells

Arterial & cardiac thrombosis

Turbulence produces counter - currents (Eddy currents) & causes endothelial injury

Brings platelets in contact with the endothelium

Induce endothelial activation

Enhance pro coagulant activity

Stasis → Stagnant blood flow

- Major contributor in the development of venous thrombi
- Promotes endothelial activation
- Enhance pro coagulant activity
- Prevents washout & dilution of activated clotting factors
- Prevents inflow of clotting factor inhibitors

Classification of thrombi

- A- according to the ***site***
- B- according to the ***color***
- C- according t presence of or absence of ***bacteria***
- D- according to whether ***occlusive*** or non occlusive thrombus

A- according to the site

Arterial

- ***Atherosclerosis***
- ***Aneurysm***
- ***Arteritis***

Cardiac

- ***Valvular***---vegetations
- ***Atrial*** ---left atrium(Mitral Stenosis)
- ***Ventricular***
  - Mural---over Myocardial infarction
  - Agonal—right v before death

Venous

- ***Phlebothrombosis***
  - In ***non*** inflamed vein (varicose vein, Heart failure, post labor or surgery)
- ***Thrombophlebitis***
  - In inflamed vein (bacteria and irradiation)

Capillary

- Rare ------frost bite

B- According to the colour

Pale (greyish white)

- ***Platelets***>fibrin
- ***grey***
- Early phase of thrombus

Red

- ***Fibrin***>platelets
- Yellowish but fibrin entangle RBC's so appear ***red***
- Veins in stasis

Mixed

- ***Platelets=fibrin***
- Alternating lamellae of ***grey and yellow***
- Arterial side of heart and veins

C - According to presence or absence of bacteria

Aseptic (Sterile)

- Thrombus containing ***no bacteria***
- ***Ex:*** caused by ***irradiation***

septic

- Thrombus containing ***bacteria***
- ***Ex:*** septic thrombophlebitis and septic vegetations in acute bacterial endocarditis

D - According to whether occlusive or not

Occlusive

- Lumen is ***occluded*** by thrombus
- Ex: ***vascular*** lumen (in blood vessel)

non occlusive

- Lumen ***not occluded*** by thrombus
- Ex: ***capacious*** site like aorta and heart (wide site)

Fate of thrombus

A- septic thrombi

fragmentation------septic emboli------pyemic abscess

B- aseptic thrombi

1-***l***ysis

2-***o***rganization

3-***r***ecanalization

4-***e***mbolization

5-***c***alcification

6-***o***cclusion

7-***i***nfection

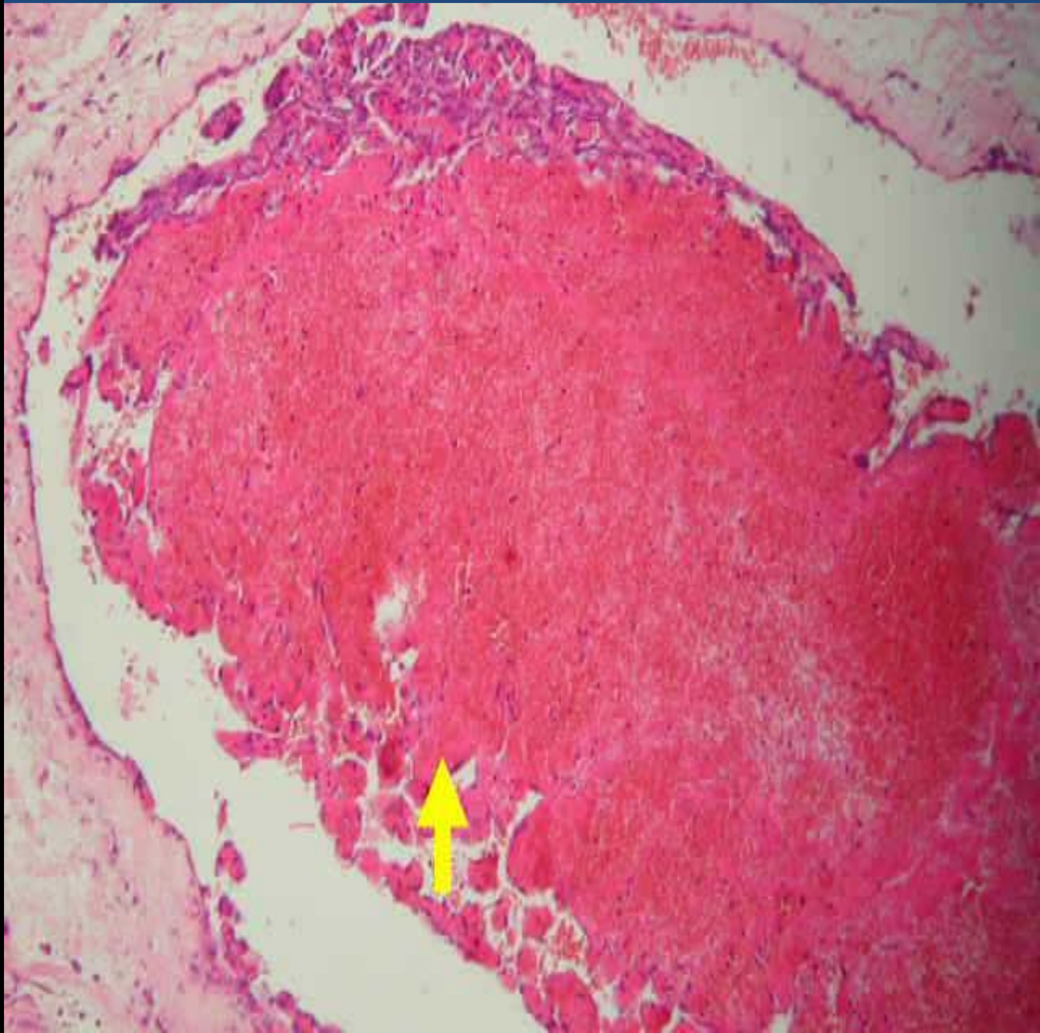
l color e

Fate of aseptic thrombi

- ***1- lysis:***

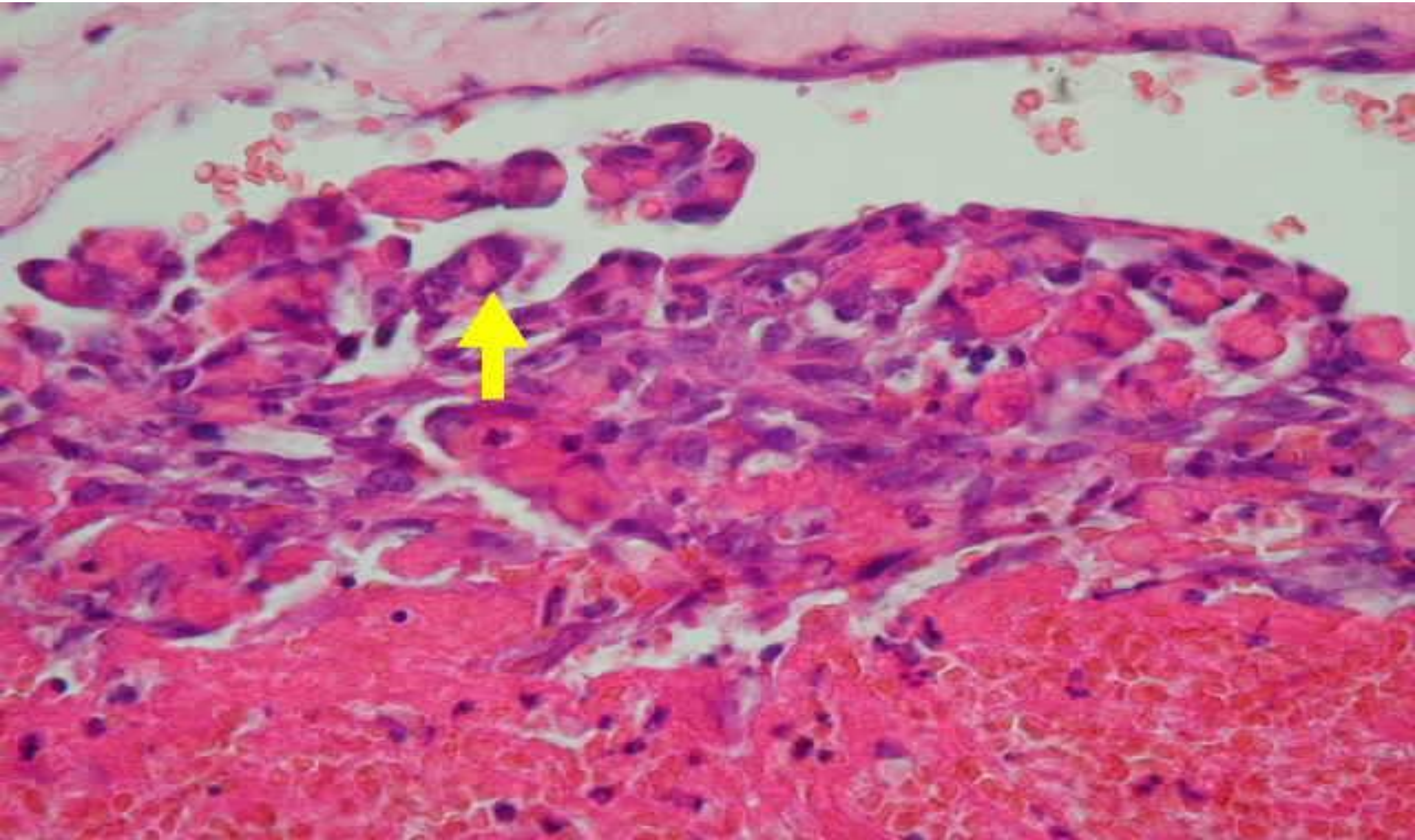
  occurs when fibrinolytic mechanisms ***break up the thrombus*** and blood flow is restored to the vessel.

2-Organization

Fresh thrombi in the dilated blood vessel (arrow).	Granulation tissue invades thrombus from subendothelial surface
	

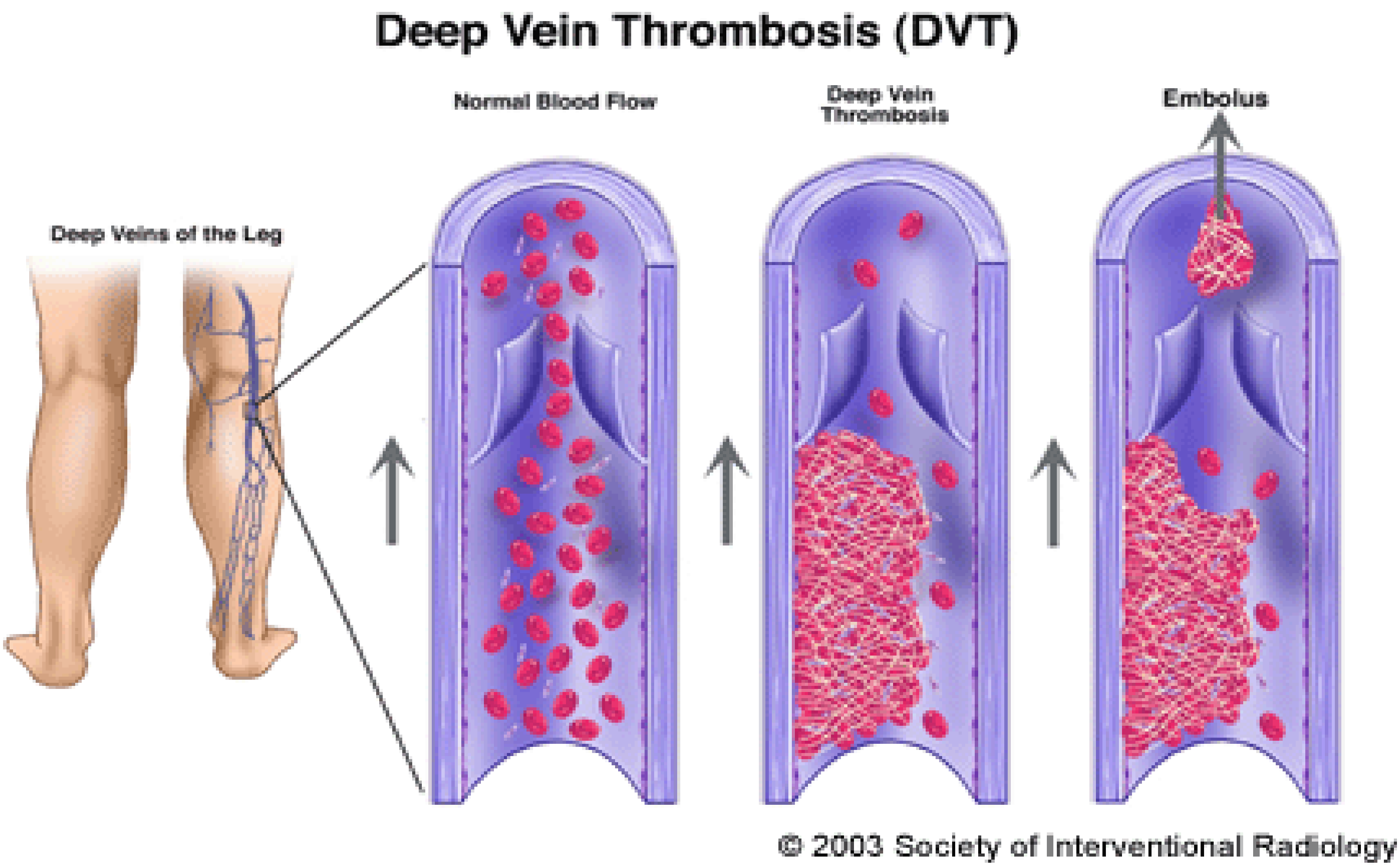
3- Recanalization

capillaries form wide anastomosing channels cross between two ends of the thrombus

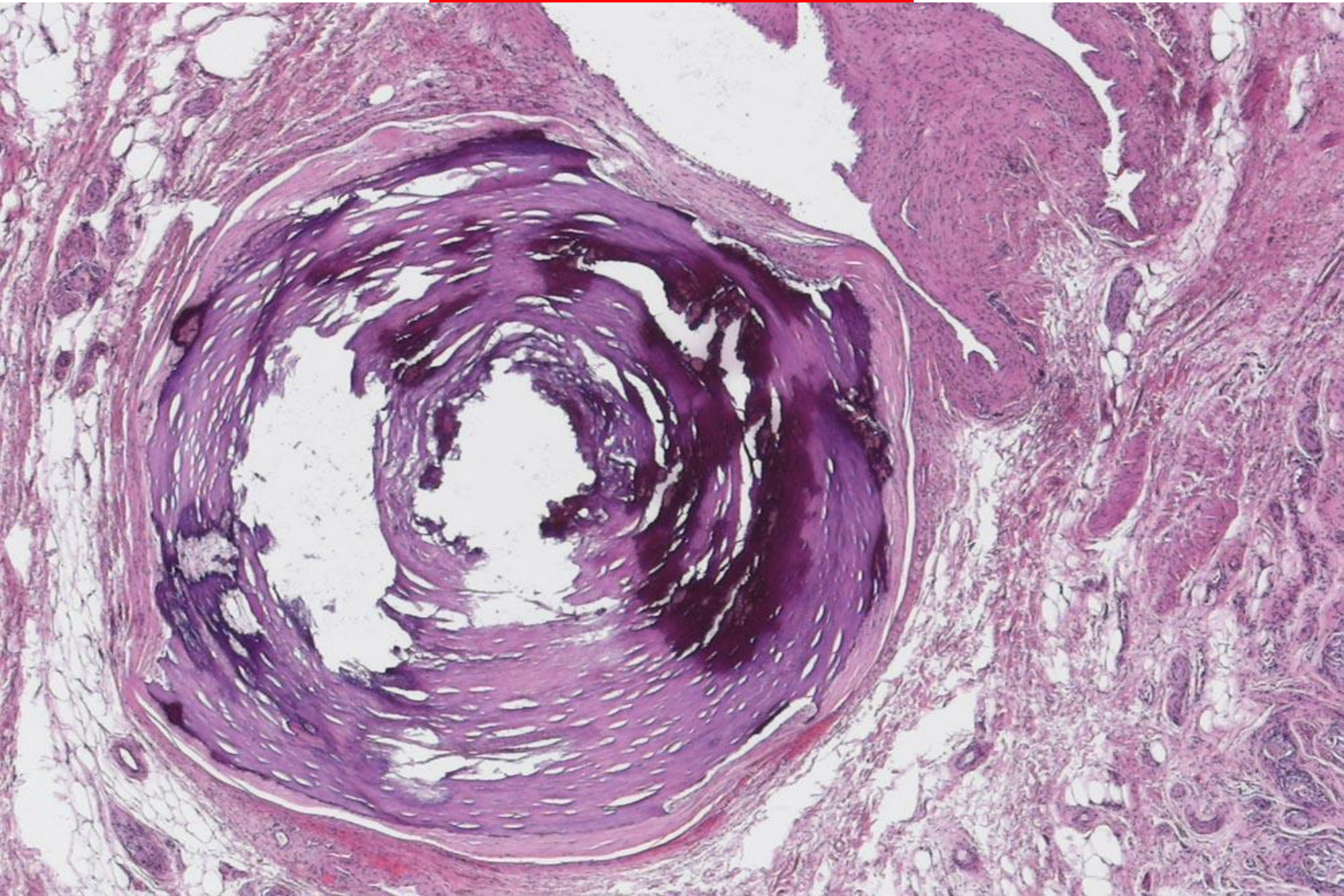


4-Embolization

Detachment of the thrombus into embolus



5-Calcification



6- Occlusion

arterial	venous	propagating
• ischemia	• Congestion and oedema	

	Thrombus	*Clot*
Time of formation	**Antemortem (during life)**	Ant-mortem or postmortem
Location	**intravascular**	Extra and intravascular
Types of blood	**Circulating**	Stagnant
Attachment to vascular wall	**Attached** at certain site	**Not attached**
Composition	**Platelets** Fibrin RBC's and WBC's	**No platelets** Fibrin RBC's and WBC's
Lines of zahn	**Present**	Absent
Consistency	**Friable**	**Elastic**
Colour	Pale Red Mixed	Red yellow

EMBOLISM

Embolus: **a detached and circulating intravascular mass that circulate in the blood to a site distant from its point of origin.**

Embolism: **impaction of the embolus in a blood vessel.**

THANK YOU